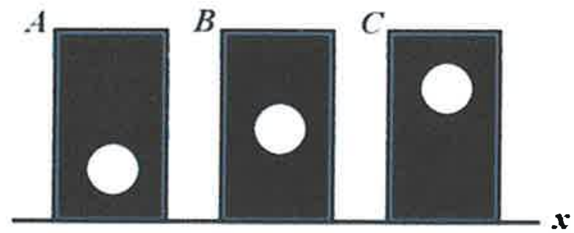
2.4 A general shape with an area of 20 $in^2$ has a moment of inertia of 106.7 $in^4$ with respect to the x axis. The centroid is located 2 inches above the x axis. What is the moment of inertia with respect to a centroidal x' axis?

$$I_x = \bar{I} + A d_y^2$$

$$\bar{I} = I_x - A y^2 = 106.7 - (20)(2)^2$$

$$\boxed{\bar{I} = 26.7 \text{ in}^4}$$

2.5 The figure shows three areas. Each area is a rectangle with a missing circle. The three rectangles have the same dimensions, and the three missing circles have the same dimensions. Which area has the largest moment of inertia about the *x* axis?




$$I_x = \int_A y^2 \, dA$$

$I_x$ for A is largest.

**Section 3 – Product of Inertia**

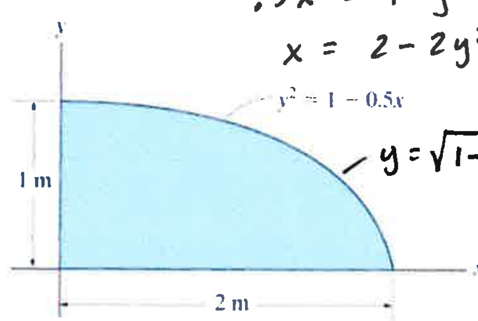
3.1 When using the parallel axis theorem to determine the product of inertia for an area about a set of non-centroidal x-y axes, $I_{xy} = \bar{I}_{x'y'} + A d_x d_y$, the second term on the right side of the equation is:

A. Area times the differential widths dx and dy
✓B. Area times the distances between the x-y origin and the x'-y' origin.
C. Always positive
D. Always negative
E. Always zero

$.5x = 1 - y^2$

$x = 2 - 2y^2$

3.2 Determine the product of inertia of the shaded area with respect to the *x* and *y* axes.



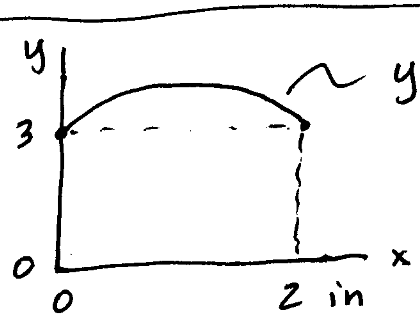

$y = \sqrt{1 - .5x}$

$$I_{xy} = \int_0^2 \int_0^{\sqrt{1-.5x}} xy \, dy \, dx$$

TI89: $\int(\int(x*y, y, 0, \sqrt{1-.5*x}), x, 0, 2)$

$$= \boxed{.333 \text{ m}^4 = I_{xy}}$$

# POI ($I_{xy}$) Integration for a Trinomial using TI-89

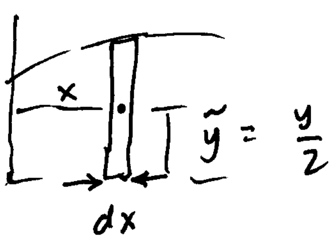


$y = [-x^2 + 2x + 3]$ ~ this is a "trinomial"

Find $I_{xy}$ for this area.

$$\boxed{I_{xy} = \int_A xy\, dA}$$

## Using a single integral:



$dA = (-x^2 + 2x + 3)\, dx$

$\tilde{x} = x$

$\tilde{y} = y/2$

$$I_{xy} = \int_0^2 x\left(\frac{-x^2 + 2x + 3}{2}\right)(-x^2 + 2x + 3)\, dx$$

To solve this by hand need to multiply (square) the trinomial... Painful!

On the TI-89: $\int(.5 * x * (-x^\wedge 2 + 2 * x + 3)^\wedge 2, x, 0, 2)$

$$\boxed{I_{xy} = 13.53 \text{ in}^4}$$

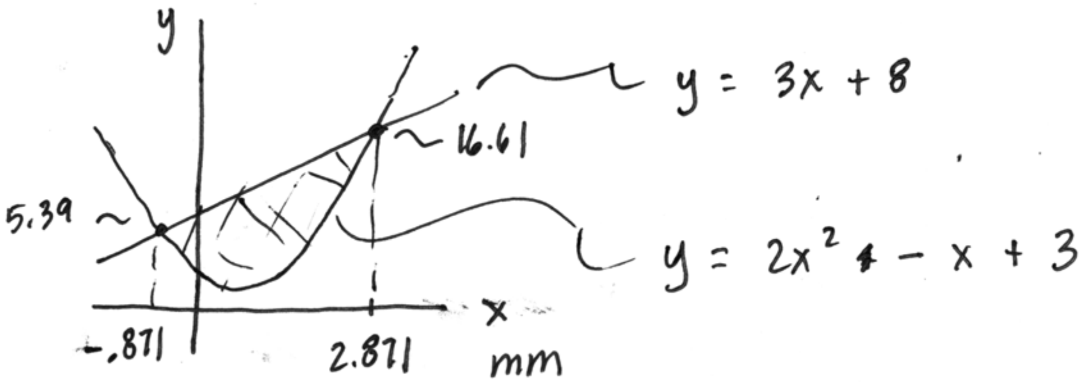
## Using a double integral:

$$I_{xy} = \int_0^2 \int_0^{-x^2 + 2x + 3} xy\, dy\, dx$$

} End up having to square it again! Painful!

TI-89: $I_{xy} = \int(\int(x * y, y, 0, -x^\wedge 2 + 2 * x + 3), x, 0, 2)$

$$\boxed{I_{xy} = 13.53 \text{ in}^4}$$




$$I_{xy} = \int_A xy \, dA$$

$$I_{xy} = \int_{-.871}^{2.871} \int_{2x^2-x+3}^{3x+8} xy \, dy \, dx$$

<u>TI-89</u>

$$I_{xy} = \int(\int(x*y,\ y,\ 2*x\text{^}2-x+3,\ 3*x+8),\ x,\ -.871,\ 2.871)$$

$$\boxed{I_{xy} = 179.85 \ \text{mm}^4}$$